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Fire-YOLO26: an NMS-free dual-stream RGB–thermal network for real-time UAV forest fire detection in smoke-challenging conditions

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Early and reliable forest fire detection is essential for ecological security and emergency response, yet UAV-based detectors often suffer from NMS-induced latency and modality-specific failures under smoke occlusion and thermal distractors. This study proposes Fire-YOLO26, an end-to-end NMS-free dual-stream RGB–thermal infrared (TIR) network built on YOLO26 for real-time UAV fire monitoring. A parallel RGB–TIR backbone preserves one-to-one matching inference, while a Thermal-guided Feature Alignment Module (T-FAM) derives a heat-driven spatial mask from TIR features to recalibrate RGB representations and suppress fire-like background noise. To enhance early small-fire learning, a Dual-modal Small Target Aware Labeling (Dual-STAL) strategy augments label assignment with thermal consistency. On the FLAME 2 multispectral UAV dataset, Fire-YOLO26-s achieves 94.6% mAP@0.5% and 74.2% mAP@0.5:0.95, improving mAP@0.5 by 6.5 points over the RGB-only YOLO26-s baseline, and raises heavy-smoke performance from 71.5% to 95.1% mAP@0.5. The model runs at 23 FPS on a Jetson Orin NX, indicating practical onboard deployment for all-weather early fire detection.

KEYWORDS
edge deployment, end-to-end object detection, forest fire detection, NMS-free, RGB–thermal fusion, small object detection, UAV, YOLO26

1 Introduction

With global climate warming, the frequency and destructive power of forest fires are increasing, posing a serious threat to ecological balance and human society. Public monitoring platforms such as Global Forest Watch provide near-real-time fire monitoring products and indicate that fire-driven forest loss remains substantial worldwide (World Resources Institute, 2024). Compared to the long revisit cycles of satellite remote sensing and the limited coverage of ground sensors, Unmanned Aerial Vehicles (UAVs) have become an indispensable component of forest fire monitoring systems due to their high mobility, low-altitude high-resolution imaging capabilities, and relatively low deployment costs (Akhloufi et al., 2021; Barmpoutis et al., 2020; Hu et al., 2025). Early UAV-based forest fire monitoring and measurement systems have also demonstrated the value of onboard sensing and automation for timely situational awareness (Merino et al., 2012).

However, actual field inspection environments are extremely complex. Early fire sources are often hidden under tree canopies or wrapped in thick smoke, and are easily confused with natural phenomena such as autumn red leaves, sunset glow, and water reflections. Existing comprehensive reviews have systematically delineated the core research gaps in wildfire detection, which are mainly concentrated in three dimensions: the contradiction between early detection accuracy and onboard real-time deployment, the performance limitation of single-modal perception in complex field scenarios, and the edge deployment dilemma arising from the difficulty of balancing accuracy and efficiency in multi-modal methods.(Das et al., 2026; Vasconcelos et al., 2024; Mohapatra and Trinh, 2022). This environmental uncertainty poses dual challenges to the robustness and real-time performance of target detection algorithms (Shamsoshoara et al., 2021). Traditional computer vision methods, such as color space segmentation algorithms (Phillips et al., 2002), perform reasonably well in simple scenarios but have extremely high false detection rates in complex backgrounds. Early probabilistic temporal modeling approaches such as hidden Markov models have also been explored for flame-related video cues, yet remain sensitive to scene variations (Toreyin et al., 2005). Methods based on hand-designed features are difficult to generalize to diverse fire morphologies and observation conditions (Sun et al., 2023).

In the field of deep learning, object detection algorithms have evolved from region-based two-stage pipelines to efficient one-stage paradigms and, more recently, end-to-end formulations. Two-stage detectors such as Faster R-CNN (Ren et al., 2015) achieve strong accuracy by generating region proposals, while one-stage detectors such as SSD (Liu et al., 2016) and RetinaNet (Lin et al., 2017) improve efficiency and address class-imbalance issues in dense prediction. For edge-friendly deployment, scalable lightweight detectors such as EfficientDet further improve the accuracy–efficiency tradeoff (Tan et al., 2020). Meanwhile, Transformer-based vision backbones, including Vision Transformer (ViT) (Dosovitskiy et al., 2021) and hierarchical Swin Transformer (Liu et al., 2021), have broadened receptive fields and enhanced global context modeling, which is beneficial under occlusion and complex backgrounds.

Traditional YOLO architectures, although highly efficient (Redmon et al., 2016; Redmon and Farhadi, 2017; Redmon and Farhadi, 2018; Jordan et al., 2024; Jocher and Qiu, 2024; Hu et al., 2022; Wang et al., 2025), heavily depend on Non-Maximum Suppression (NMS) to filter redundant bounding boxes, leading to unacceptable latency variance in dense forest fire scenarios. Similarly, the DETR architecture (Carion et al., 2020; Zhu et al., 2021) pioneered end-to-end detection but often required computationally expensive overhead. Wang et al. (2024) pioneered the consistent dual assignments strategy in YOLOv10, marking the YOLO family's first successful NMS-free deployment (Wang et al., 2024). Subsequently, Tian et al. (2025) introduced an attention-centric design in YOLOv12 to improve feature representation (Tian et al., 2025). Building upon these innovations, the latest YOLO26 architecture (Sapkota et al., 2025; Chakrabarty, 2026) adopts a pure end-to-end "one-to-one" matching mechanism and removes the Distribution Focal Loss (DFL) layer, which can hinder inference speed on edge devices, making it well-suited for UAV deployment.

However, directly applying generic YOLO26 to forest fire detection still has limitations. Generic models mainly rely on visible-light (RGB) images, while in the early stages of fire, smoke is often visually captured before open flames, causing severe line-of-sight occlusion. Although thermal infrared (TIR) imaging has smoke-penetration and temperature anomaly cues, its lack of texture and lower spatial detail make single-modality infrared detection prone to misclassifying heated rocks, vehicle exhaust, or sunlit surfaces as fire sources. Moreover, early-stage fire targets are often extremely small and partially occluded, which is a known challenge in small object detection (Tong et al., 2020).

To overcome the physical limitations of single-modality perception and fully leverage the advantages of YOLO26's advanced end-to-end inference architecture, this paper proposes the Fire-YOLO26 network. Unlike previous studies that used simple image overlay (Frizzi et al., 2016) or computationally expensive load decision-level fusion (Muhammad et al., 2019), this work focuses on achieving lightweight, deep multi-modal fusion at the intermediate layers of feature extraction (Liu K. et al., 2025; Zhang et al., 2023).

Specifically, we design a parallel dual-stream CSP-based backbone to process RGB and thermal infrared (TIR) inputs independently: the RGB stream extracts texture, color and geometric features, while the TIR stream captures temperature anomaly and heat source contour features. We embed the proposed Thermal-guided Feature Alignment Module (T-FAM) at the P3, P4, and P5 output layers of the dual backbones (the three core intermediate layers of the feature extraction network) to perform progressive modal fusion, which preserves the end-to-end one-to-one matching inference property of YOLO26 while achieving deep feature complementarity.

Using the strictly aligned RGB-TIR image pairs from the FLAME 2 dataset (Hopkins et al., 2022), we train this dual-stream network. Quantitative results verify the lightweight property of our fusion strategy: the smallest variant Fire-YOLO26-n achieves a model size of 7.6 M parameters, a computational cost of 18.4 G FLOPs, and a single-frame processing latency of 24.8 ms on RTX 4090 and 42.7 ms on the edge device Jetson Orin NX. Compared with the mainstream single-modal YOLO11s detector, our method reduces inference latency by 3.5% while improving mAP@0.5 by 12.2%, achieving a superior accuracy-efficiency tradeoff. The main contributions are as follows:

1. We propose a YOLO26-based dual-stream forest fire detection network, Fire-YOLO26, achieving deep end-to-end fusion of RGB and thermal infrared features;
2. We design a Thermal-guided Feature Alignment Module (T-FAM) that utilizes infrared thermal radiation information to dynamically guide visible-light feature focusing, effectively suppressing fire-like background noise;
3. We propose a Dual-modal Small Target Aware Labeling strategy (Dual-STAL) to enhance the model's detection capability for early tiny high-heat targets;
4. We validate the effectiveness of the method on the FLAME 2 dataset and achieve real-time inference on edge devices that meets practical requirements.

In summary, this study addresses three critical research gaps in existing UAV-based forest fire detection methods: the modal-specific failure of single-modality detectors under smoke occlusion and thermal distractors, the loss of early tiny fire targets, and the contradiction between detection accuracy and real-time inference latency for onboard edge deployment. To bridge these gaps, we propose the Fire-YOLO26 network with three core innovations: a dual-stream end-to-end fusion architecture compatible with YOLO26's NMS-free design, a T-FAM module for thermal-guided feature alignment and noise suppression, and a Dual-STAL strategy for early small fire enhancement. The rest of this manuscript is organized as follows: Section 2 describes the dataset, preprocessing pipeline, the overall architecture of Fire-YOLO26, and the core innovation modules; Section 3 presents the experimental setup, comparison results, ablation studies, and visualization analysis; Section 4 discusses the advantages, limitations, and future directions of the proposed method; Section 5 summarizes the main conclusions of this study.

2 Methods

2.1 Dataset and preprocessing

This study uses the highly authoritative FLAME 2 (Fire detection and modeLing: Aerial Multi-spectral imagE) dataset (Hopkins et al., 2022) as the experimental benchmark. Released in 2022, this dataset contains synchronized RGB and thermal infrared video pairs collected by a DJI Mavic Enterprise Advanced UAV during a prescribed burn in Northern Arizona. It provides well-aligned RGB-thermal image pairs under smoke and complex fire scenarios. The dataset includes both controlled burn experiments, with annotations for fire and smoke categories. Compared with the earlier FLAME dataset (Shamsoshoara et al., 2021), as well as other general fire datasets like D-Fire (de Venâncio et al., 2022) and DetectiumFire (Liu Z. et al., 2025), FLAME 2 provides more accurate frame-level alignment and richer samples of fire morphologies including open flames and smoldering.

In the data preprocessing stage, to eliminate parallax caused by differences in camera mounting positions, we use a Homography Matrix to perform strict pixel-level registration of RGB and thermal infrared images. The registration process employs SIFT feature point matching and RANSAC algorithm to estimate the transformation matrix, achieving sub-pixel registration accuracy. Additionally, considering that infrared images are typically single-channel radiation data, we apply Min-Max normalization and map them to a three-channel tensor format matching the RGB feature space. The dataset is divided into training, validation, and test sets in an 8:1:1 ratio.

To enhance the model's generalization capability for small targets, Mosaic (Bochkovskiy et al., 2020), Mixup (Zhang et al., 2018), and CutMix (Yun et al., 2019) strategies are employed during training. Mosaic enhancement increases the number and diversity of small targets by randomly stitching four images; Mixup improves model robustness by linearly interpolating images and labels. Furthermore, we introduce data augmentation techniques including random flipping, random rotation (± 15°), random scaling (0.5–1.5x), and color jittering.

2.2 Network architecture

The overall architecture of Fire-YOLO26 follows the principles of "dual-stream perception, feature complementarity, and end-to-end output," as shown in Figure 1. The network abandons traditional multi-stage processing and adopts parallel dual-stream backbone networks to process RGB and infrared inputs separately.

The backbone follows the Cross Stage Partial (CSP) design philosophy, which has been widely used to improve feature learning efficiency and reduce computation (Wang et al., 2020; Sapkota et al., 2025), but reduces the base channel count of single-stream branches from the default 80 to 64 through pruning techniques, ensuring that the total computational load remains suitable for edge deployment when both streams run in parallel. The RGB stream focuses on capturing flame color features, smoke texture details, and environmental background geometric structures; the infrared stream focuses on extracting contours and intensity information of temperature anomaly regions. The two branches interact and fuse through the T-FAM module during the feature extraction phase, eventually converging at YOLO26's Decoupled Head, directly outputting target coordinates and category confidence without redundant boxes, achieving end-to-end inference.

Fire-YOLO26's detection head adopts an NMS-Free design, where each grid point predicts only one optimal target box. Through a one-to-one label assignment strategy, positive samples are determined during the training phase, thereby completely eliminating the time-consuming computationally expensive post-processing steps found in traditional detectors. This design has significant engineering advantages in dense fire point scenarios.

2.3 Thermal-guided feature alignment module

In the multi-modal fusion process, the most significant challenge lies in the semantic misalignment of features from different modalities. For example, in heavy smoke scenarios, fire sources in RGB images may be completely occluded while being clearly visible in infrared images; conversely, under strong sunlight, hot rocks may produce strong responses in infrared images while lacking visual flame features in RGB. Simple feature addition can lead to RGB noise contaminating effective infrared information, or *vice versa*.

To address this, we propose the Thermal-guided Feature Alignment Module (T-FAM), whose structure is shown in Figure 2.

The core insight behind T-FAM is that thermal infrared provides reliable cues about where potential fire sources exist, even when smoke obscures the visible appearance. T-FAM leverages this "heat map" as a spatial guidance signal to selectively enhance RGB features in high-temperature regions while suppressing them in irrelevant background areas. This mimics human visual attention: when searching for fires, observers first detect heat anomalies (thermal) and then verify fire characteristics (RGB), rather than processing both modalities independently.

Rather than directly fusing features, this module first calculates a spatial attention map from the infrared feature map, generating a dynamic "Heat Mask." The corresponding calculations are formally

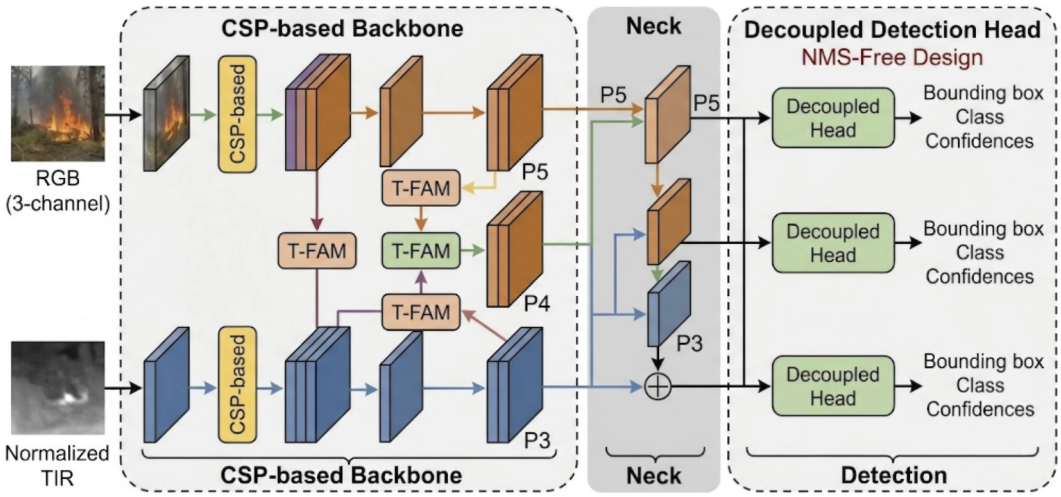


FIGURE 1
Overall architecture of the Fire-YOLO26 network.

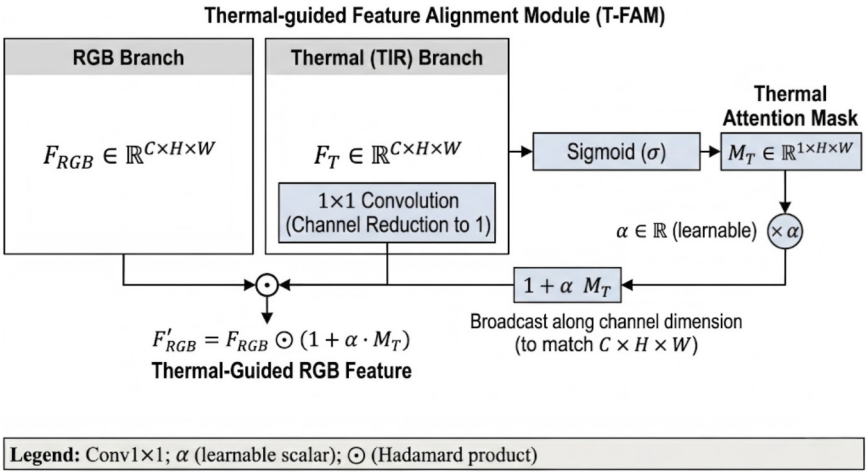


FIGURE 2
Structure diagram of the Thermal-guided Feature Alignment Module (T-FAM).

defined in Equations 1–3. Specifically, given the feature map from the infrared branch $F_T \in \mathbb{R}^{C\times H\times W}$, we first compress it to a single channel through 1×1 convolution, then generate spatial attention weights through the Sigmoid activation function, inspired by foundational spatial attention mechanisms (Woo et al., 2018):

$$M_T = \sigma(W_c * F_T + b) \tag{1}$$

where σ denotes the Sigmoid function, and W_c and b are learnable convolution parameters. This mask $M_T \in \mathbb{R}^{1\times H\times W}$ characterizes high-confidence heat source regions in the field of view.

Subsequently, T-FAM uses this mask to perform spatially weighted correction on the RGB feature map $F_{RGB} \in \mathbb{R}^{C\times H\times W}$:

$$F'_{RGB} = F_{RGB} \odot (1 + \alpha \cdot M_T) \tag{2}$$

where α is a learnable scaling factor and $\odot$ denotes element-wise multiplication. In high-temperature regions, the network is forced to focus on subtle texture variations in RGB; in low-temperature regions (such as background forests), RGB feature responses are suppressed. This mechanism simulates the human cognitive process of first perceiving heat sources and then confirming fire situations, effectively solving the problem of target feature blurring caused by smoke occlusion and background interference. In contrast to recent state-of-the-art multi-modal fusion frameworks, such as the Multidimensional Representation Collaborative Fusion (MRCF) in MCDet (Xu et al., 2025) which relies on complex global interaction, or the frequency-domain spectral filtering introduced in LF-DETR (Qi et al., 2026), our T-FAM prioritizes computational efficiency. By leveraging a unidirectional spatial heat-guidance mechanism, T-FAM avoids the high overhead of global attention

matrices and remains aligned with YOLO26's edge-first design philosophy.

2.4 Dual-modal small target aware labeling

The Small Target Aware Labeling (STAL) strategy introduced by YOLO26 (Sapkota et al., 2025), while performing excellently on general datasets, still has room for improvement when dealing with forest fires. While slicing-aided methods like SAHI (Akyon et al., 2022) have shown promise for general small object detection, they often incur significant computational overhead during inference due to the overlapping patch processing. In contrast, this paper proposes an enhanced Dual-STAL strategy that operates inherently within the network's labeling process.

Traditional label assignment relies solely on geometric overlap (IoU), which is problematic for small early-stage fires: a predicted box may slightly miss the tiny fire region (low IoU) while still covering the correct thermal center. Dual-STAL addresses this by incorporating thermal confidence,if a prediction captures high-temperature pixels in the infrared image, it should receive positive supervision even with imperfect spatial alignment. This dual-criterion quality estimation ensures that thermally-relevant predictions are preserved during training.

Dual-STAL introduces a thermal consistency constraint. Specifically, we define a dual-modal quality estimation function:

$$Q = w_1 \cdot IoU + w_2 \cdot C_T \tag{3}$$

where $C_T \in [0, 1]$ represents the normalized temperature confidence in the predicted box region, computed as the average infrared intensity within the box divided by the maximum intensity in the thermal image. The weight coefficients $w_1 = 0.6$ and $w_2 = 0.4$ are determined through ablation experiments (see Supplementary Material), balancing geometric accuracy and thermal relevance. Only when $Q > \theta$ (threshold $\theta = 0.5$) is a sample classified as a high-quality positive sample.

This strategy forces the network to consider both visual boundary and thermal radiation boundary consistency when regressing small target bounding boxes, thereby significantly improving recall for distant early fire points with extremely small pixel occupancy (less than 32 × 32 pixels), which is consistent with the common definition and challenges of small object detection in the literature (Tong et al., 2020). Furthermore, to overcome the gradient oscillation typically associated with sparse positive samples in NMS-free architectures, our training process leverages the MuSGD optimizer (Jordan et al., 2024) natively supported by YOLO26. MuSGD derives from the Muon optimizer utilized in large-scale neural networks, which performs semi-orthogonalization on gradient matrices to achieve steepest descent under the spectral norm. This ensures stable gradient backpropagation for the highly non-linear multi-modal feature space. Dual-STAL operates synchronously with YOLO26's Progressive Loss Balancing (ProgLoss) mechanism (Chakrabarty, 2026), ensuring that the dynamic loss weights protect the gradient contribution of the hard-to-detect small fire targets identified by Dual-STAL during the later stages of training.

2.5 Hardware system and deployment environment

To verify the training efficiency and onboard real-time deployment capability of the proposed Fire-YOLO26 network, we built a two-stage hardware system covering model training and UAV edge inference, with the system-level architecture shown in Figure 3. The specific hardware components and selection justification are detailed as follows:

2.5.1 Model training hardware platform

The training platform is built on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU, which has 24 GB of GDDR6X video memory, 16384 CUDA cores, and 512 Tensor cores. The workstation is configured with an Intel Core i9-13900K processor, 64 GB DDR5 memory, and 2 TB NVMe solid-state drive. The software environment includes CUDA 12.1, cuDNN 8.9, and PyTorch 2.1.0. The RTX 4090 GPU provides sufficient parallel computing power for the training of the dual-stream multi-modal network, and its large-capacity video memory supports a batch size of 16 at 640 × 640 input resolution, ensuring stable and efficient model training within 300 epochs.

2.5.2 UAV onboard edge inference hardware platform

The edge deployment platform uses the NVIDIA Jetson Orin NX embedded system-on-module (SOM), which is the core onboard computing unit for UAVs. It integrates a 6-core ARM Cortex-A78AE CPU, a 384-core NVIDIA Ampere architecture GPU, and 16 GB 128-bit LPDDR5 shared memory with a maximum AI computing power of 100 TOPS. The UAV platform used for data acquisition in the FLAME 2 dataset is the DJI Mavic Enterprise Advanced, which is equipped with a 1/2-inch CMOS RGB sensor (20 million effective pixels) and a 640× 512 resolution uncooled thermal infrared camera, supporting pixel-level synchronized acquisition of RGB and TIR images.The Jetson Orin NX has the characteristics of small size, light weight, low power consumption (typical power consumption of 10–15W during inference), and high edge computing power, which perfectly matches the strict payload, power supply, and real-time inference requirements of UAV onboard deployment. Its 100 TOPS AI computing power can support Fire-YOLO26-n to achieve 23 FPS real-time inference, which fully meets the 20 FPS minimum real-time requirement of the UAV flight control closed-loop system for fire monitoring. Meanwhile, the DJI Mavic Enterprise Advanced's dual-sensor configuration provides strictly synchronized RGB-TIR image pairs, which is the data basis for the training and validation of our dual-modal network.

3 Results

To comprehensively evaluate Fire-YOLO26's performance, we conducted extensive comparison experiments and ablation studies on the FLAME 2 dataset. The experimental environment is based on NVIDIA GeForce RTX 4090 GPU (24GB VRAM),

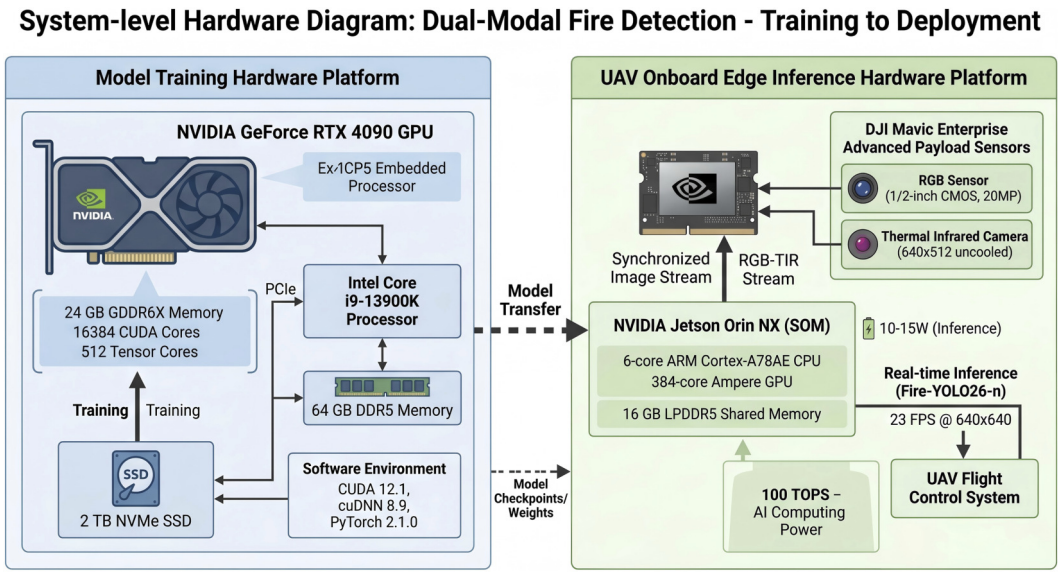


FIGURE 3
System-level hardware diagram.

TABLE 1 Performance comparison of different methods on the FLAME 2 dataset.

Method	Modality	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Params (M)	FLOPs (G)	Latency (ms)
YOLOv8n	RGB	78.4	52.3	3.2	8.7	18.5
YOLOv8s	RGB	82.1	56.8	11.2	28.6	28.3
YOLO11n	RGB	80.7	54.1	2.6	7.4	16.2
YOLO11s	RGB	84.3	58.6	9.4	25.8	25.7
YOLO26-n	RGB	85.2	59.4	3.8	9.2	12.4
YOLO26-s	RGB	88.1	63.7	12.5	31.4	21.6
YOLO26-n	TIR	81.6	55.2	3.8	9.2	12.4
YOLO26-s	TIR	85.4	60.1	12.5	31.4	21.6
Fire-YOLO26-n	RGB + TIR	**91.1**	**68.3**	**7.6**	**18.4**	**24.8**
Fire-YOLO26-s	RGB + TIR	**94.6**	**74.2**	**25.0**	**62.8**	**38.5**
Fire-YOLO26-n (Jetson)	RGB + TIR	**91.1**	**68.3**	**7.6**	**18.4**	**42.7**
Fire-YOLO26-s (Jetson)	RGB + TIR	**94.6**	**74.2**	**25.0**	**62.8**	**68.3**

Bold values indicate the best performance among all compared methods.

CUDA 12.1, PyTorch 2.1.0. Input image resolution is uniformly adjusted to 640 × 640. Training runs for 300 epochs with a batch size of 16, initial learning rate of 0.01, using Cosine Annealing learning rate scheduling strategy. We also tested inference speed on the NVIDIA Jetson Orin NX (16GB shared memory) embedded development board to evaluate edge deployment performance.

3.1 Comparison experiments

We compared Fire-YOLO26 with current mainstream single-modality and multi-modality detection algorithms, including YOLOv8 (Jocher et al., 2023), YOLO11 (Jocher and Qiu, 2024), and the original YOLO26 (Jocher and Qiu, 2024; Sapkota et al., 2025). To ensure fair comparison, all models were trained and tested

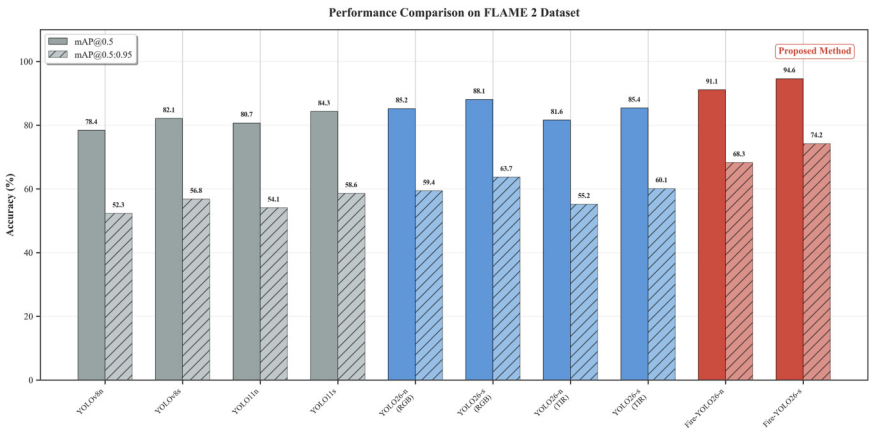


FIGURE 4
Performance comparison of different methods on the FLAME 2 dataset.

TABLE 2 **Detection performance comparison under different scenarios (mAP@0.5%).**

Method	Daytime	Heavy smoke	Average
YOLO26-s (RGB)	94.2	71.5	88.1
YOLO26-s (TIR)	89.8	82.1	85.4
Fire-YOLO26-s	**96.3**	**95.1**	**94.6**

Bold values indicate the best performance among all compared methods.

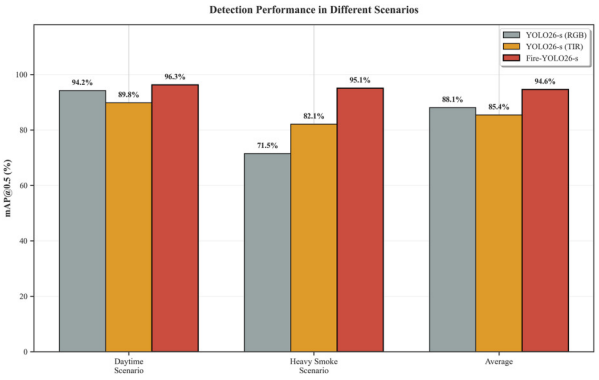


FIGURE 5
Detection performance comparison in different scenarios.

on the FLAME 2 dataset using identical training settings. Detailed quantitative results are presented in Table 1.

Experimental results show that on the full test set, Fire-YOLO26-s achieves 94.6% mAP@0.5% and 74.2% mAP@0.5:0.95, representing improvements of **6.5** percentage points over the RGB-only YOLO26-s baseline and **9.4** percentage points over YOLO26-n. Figure 4 provides a visual comparison of different methods.

From Table 2, we can see that in heavy smoke scenarios, Fire-YOLO26 improves by 23.6 percentage points over RGB single-modality, fully demonstrating the superiority of multi-modal fusion in complex environments. Figure 5 illustrates the performance comparison across different scenarios.

Furthermore, thanks to the NMS-Free architecture advantage, Fire-YOLO26-n achieves an inference latency of only **24.8 ms** (approximately 40 FPS) on RTX 4090 and 42.7 ms (approximately 23 FPS) on Jetson Orin NX, significantly outperforming traditional NMS-based detectors (e.g., YOLOv8 variants require an additional post-processing step for NMS), fully meeting the real-time requirements of UAV flight control loops. Figure 6 shows the precision-speed tradeoff analysis.

3.2 Ablation study

To verify the contribution of each innovation module, we designed a progressive ablation experiment, with results shown in Table 3. The baseline model uses an input concatenation strategy, directly stitching RGB and TIR images in the channel dimension before feeding them into a single-stream network.

Experimental data shows that replacing simple input fusion with the proposed T-FAM feature fusion module reduces the model's false positive rate from 12.3% to 6.8% (a 44.7% reduction), with mAP@0.5 improving by 3.6 percentage points, proving the effectiveness of the feature alignment mechanism in suppressing background noise. Further introducing the Dual-STAL strategy improves the model's recall for extremely small fire points (area less than 32 × 32 pixels) from 70.1% to 81.7% (a 16.6% improvement), indicating that label assignment optimization targeting physical properties is crucial for early fire detection. The ablation study results are visualized in Figure 7.

3.3 Visualization analysis

We use Grad-CAM (Selvaraju et al., 2017) to generate class activation heatmaps for visualizing model attention distribution.

From Figure 8, when processing complex dense smoke scenes with various distractors, the baseline models exhibit clear limitations. The baseline RGB model's attention (Grad-CAM) diverges significantly across the background, resulting in missed detections for tiny fires and false positives triggered by visually

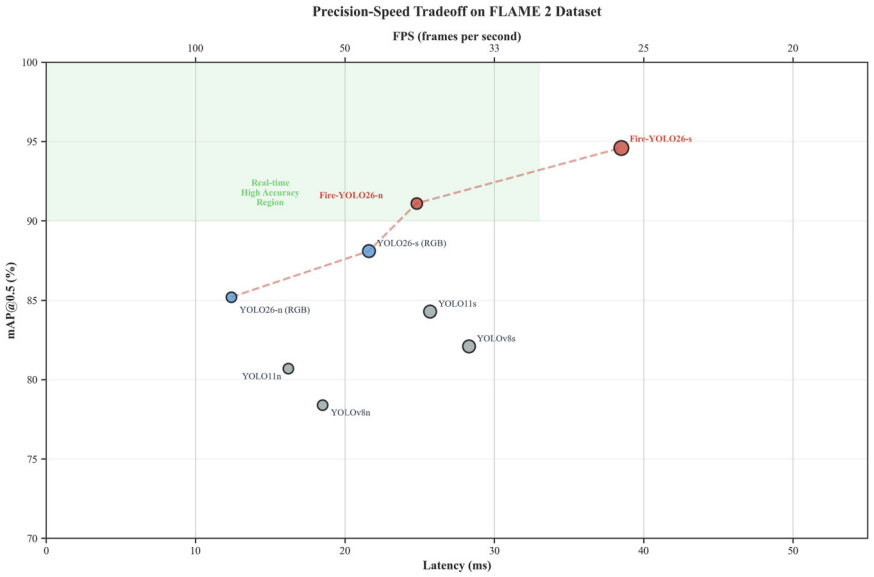


FIGURE 6
Precision-speed tradeoff comparison of different methods on the FLAME 2 dataset.

TABLE 3 Ablation experiment results.

Configuration	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Small target Recall (%)	False Positive (%)
Baseline (input Concat)	89.2	67.5	62.4	12.3
+ T-FAM	92.8	71.4	70.1	**6.8**
+ Dual-STAL	**94.6**	**74.2**	**81.7**	7.2

Bold values indicate the best performance among all compared methods.

similar objects. Similarly, the baseline TIR model is easily misled by thermal background noise and fails to capture low-resolution targets. In contrast, the Grad-CAM visualization for the proposed Fire-YOLO26-s demonstrates that its activation heat zone precisely focuses on the true fire source cores. Even when flames are completely obscured by heavy smoke, our model successfully detects the obscured tiny fires and achieves accurately aligned boundaries. This effectively proves that by fusing multi-modal features, Fire-YOLO26-s overcomes both visual and thermal interference, demonstrating superior anti-interference capability and robustness.

Figure 9 shows Fire-YOLO26's detection results in various typical scenarios:(a) clear daytime fire, (b) dense smoke obscuration, (c) distractor suppression, and (d) tiny fire source detection. It is observed that the model accurately locates obvious flames while demonstrating remarkable robustness in complex environments. Specifically, the model effectively suppresses background noise and avoids false positives even when faced with fire-like distractors (such as red leaves synthesized onto the original images). Furthermore, it maintains high-confidence detection when the fire source is extremely small or heavily obscured by thick smoke.

To provide a comprehensive comparison of model complexity, Figure 10 illustrates the parameter count and computational requirements of different methods.

Furthermore, to evaluate the onboard deployment feasibility for UAV applications, we conducted a comprehensive deployment performance test on the NVIDIA Jetson Orin NX embedded platform, which is the mainstream onboard computing unit for industrial UAVs, covering computational efficiency, power consumption, and scalability, with detailed results shown in Figure 11.

In terms of computational efficiency, Fire-YOLO26-n achieves a single-frame inference latency of 42.7 ms (23.4 FPS) and Fire-YOLO26-s achieves 68.3 ms (14.6 FPS) at 640× 640 input resolution on the Jetson Orin NX with 10W mode (the common power mode for UAV long-endurance flight). Both variants meet the 10 FPS minimum real-time requirement for forest fire monitoring, and the Fire-YOLO26-n variant can fully satisfy the 20 FPS requirement of the UAV flight control closed-loop system for dynamic fire tracking.

In terms of power consumption, the Jetson Orin NX maintains a typical power consumption of 10–12W during the continuous inference of Fire-YOLO26-n, which is only 1/45 of the power

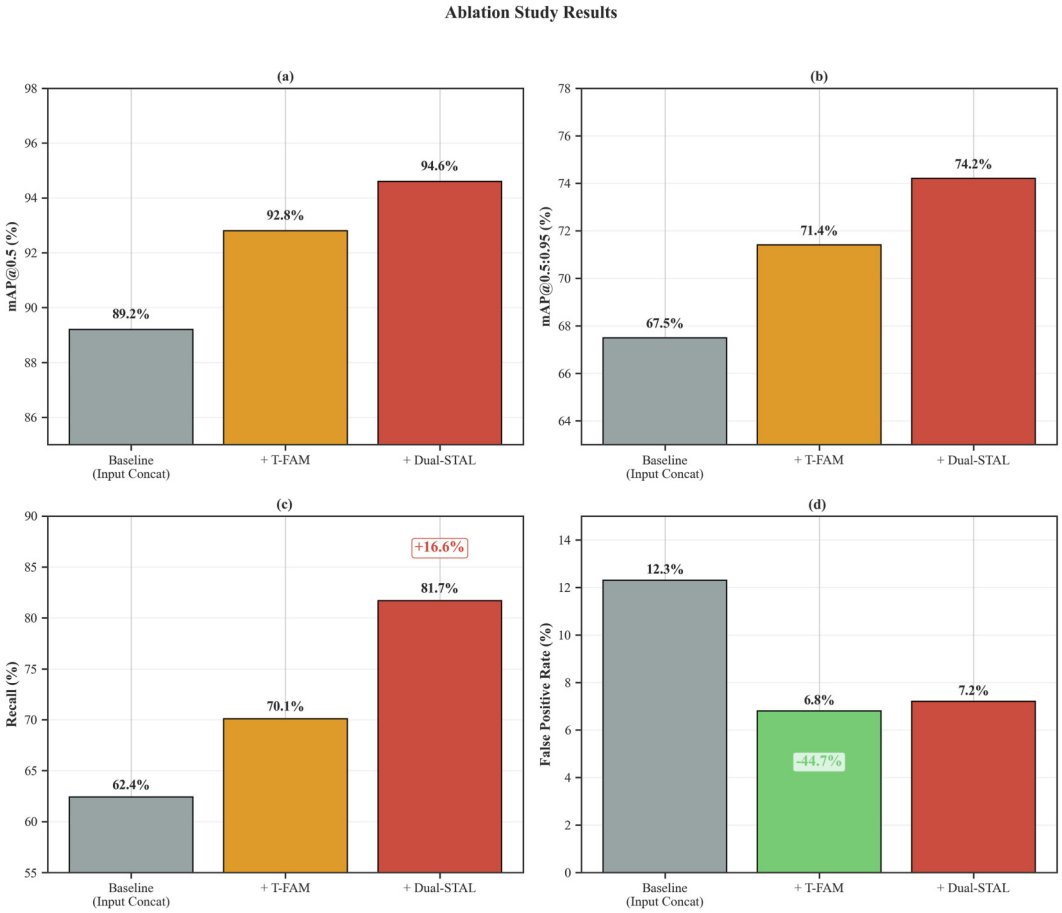


FIGURE 7
Ablation study results showing the contribution of each proposed module. **(a)** Detection accuracy. **(b)** Strict detection accuracy. **(C)** Small target recall. **(d)** False positive rate.

consumption of the RTX 4090 GPU, and perfectly matches the 3S-6S lithium battery power supply system of most medium and small industrial UAVs. Combined with the 200-300 Wh battery capacity of mainstream UAVs, the continuous working time of the onboard computing unit can reach more than 2 h, which fully meets the requirements of large-area forest patrol missions.

In terms of scalability, the proposed Fire-YOLO26 network has excellent scalability for UAV platforms with different computing power: the lightweight Fire-YOLO26-n variant can be deployed on micro-UAVs with low-power embedded chips after INT8 quantization; the high-precision Fire-YOLO26-s variant can be deployed on large UAVs with higher computing power to achieve more accurate fire and smoke segmentation. Meanwhile, the network supports input resolutions from 320× 320 to 1280× 1280, and can adaptively adjust the inference speed and accuracy according to the UAV flight altitude and patrol mission requirements.

4 Discussion

This paper proposes Fire-YOLO26, an RGB-infrared dual-stream end-to-end detection network based on the YOLO26 architecture for UAV forest fire inspection tasks, which systematically addresses four key research gaps in the field through targeted technical designs:

1. For the gap of insufficient robustness of existing multi-modal fusion methods in complex scenes: we design a lightweight dual-stream backbone with T-FAM module, which uses thermal heat masks to dynamically recalibrate RGB features, achieving deep feature-level fusion while avoiding modal misalignment and computationally expensive overhead of traditional fusion methods;
2. For the gap of severe performance degradation of detectors under heavy smoke occlusion: we leverage the smoke-penetration property of thermal infrared imaging, and use TIR-derived heat masks to guide the network to focus on the real fire source core even when the flame is completely obscured by smoke, improving the heavy-smoke scenario mAP@0.5 from 71.5% (RGB-only baseline) to 95.1%;
3. For the gap of high missed detection rate of early tiny fire targets in NMS-free architectures: we propose the Dual-STAL strategy, which introduces thermal consistency constraints into the label assignment process, improving the recall rate of tiny fire targets smaller than

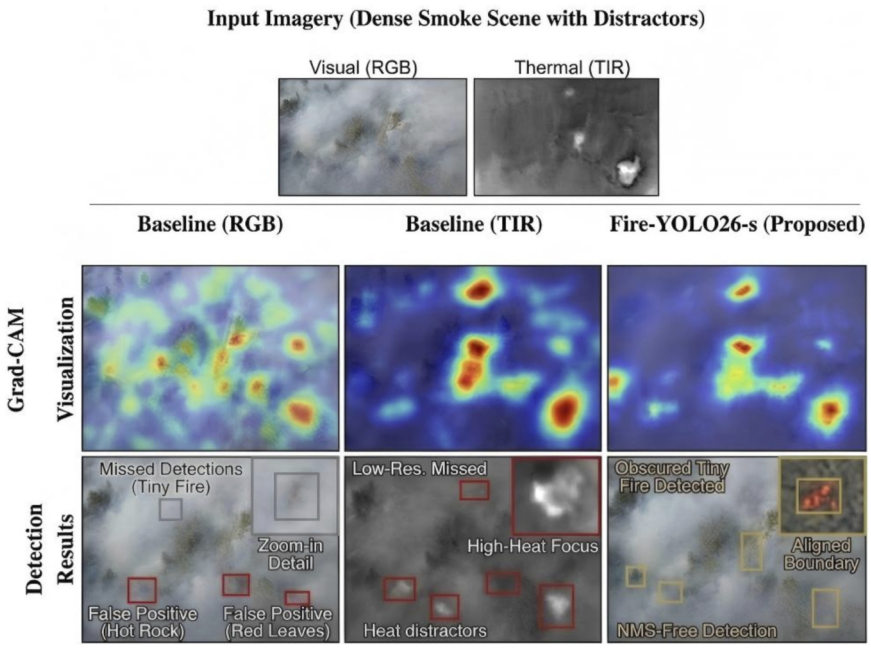


FIGURE 8
Grad-CAM visualization comparison of detection results.

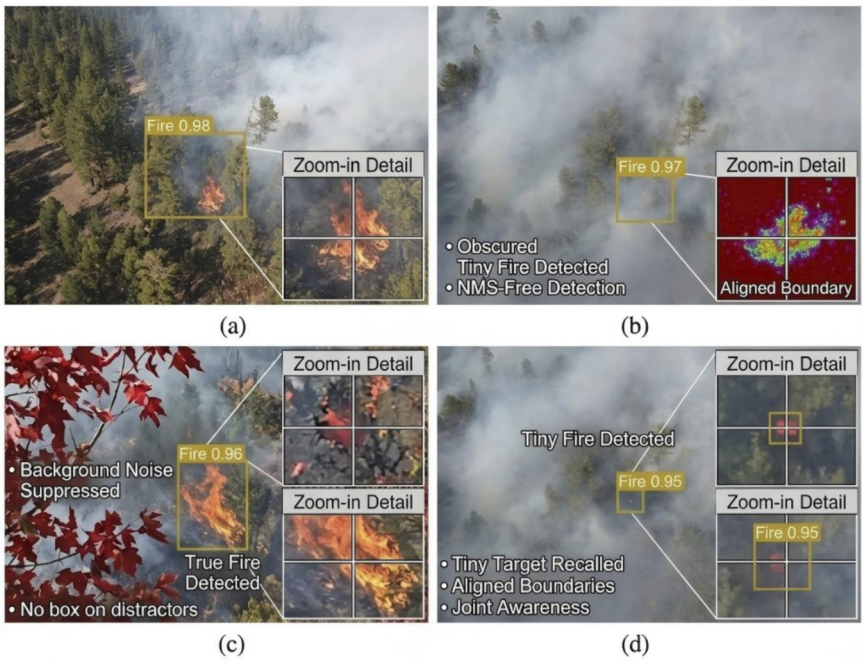


FIGURE 9
Detection result examples in typical scenarios. **(a)** Daytime clear fire. **(b)** Dense smoke obscuration. **(c)** Distractor suppression. **(d)** Tiny fire source detection.

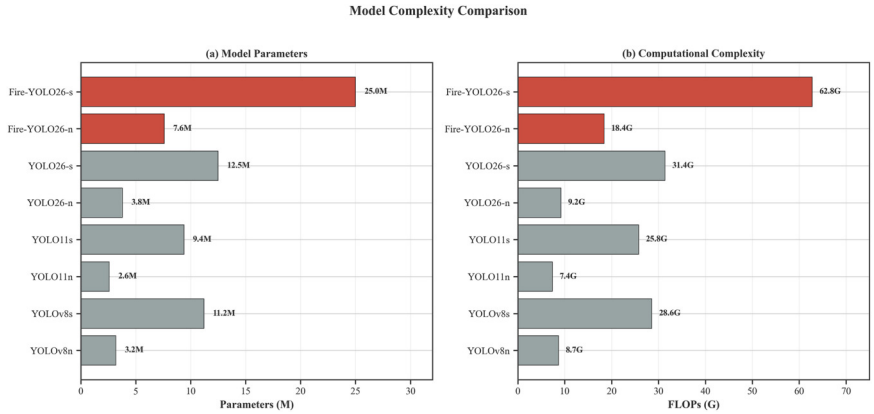


FIGURE 10
Model complexity comparison showing parameters and FLOPs for different detection methods.

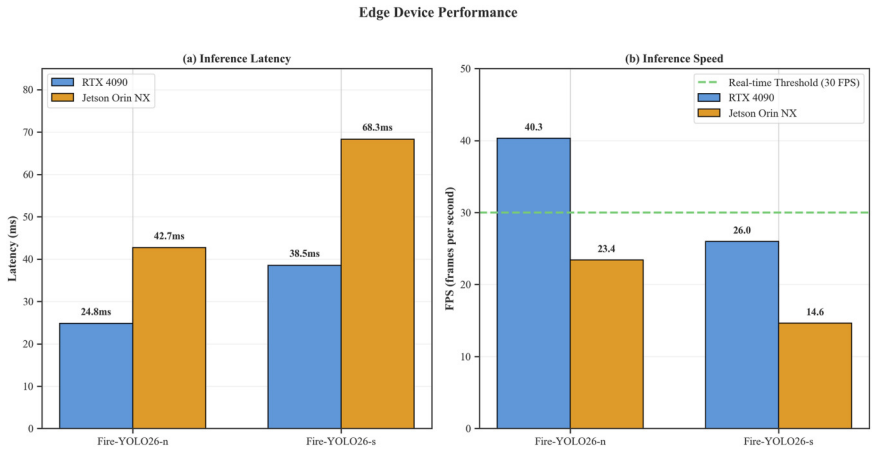


FIGURE 11
Edge device performance comparison between RTX 4090 and Jetson Orin NX.

32×32 pixels from 70.1% to 81.7%, effectively enhancing the early fire detection capability;

4. For the gap of contradiction between detection accuracy and onboard real-time deployment: we build the network based on the NMS-free YOLO26 architecture, eliminating the latency jitter caused by NMS post-processing, and achieving 23 FPS real-time inference on the Jetson Orin NX edge device, which fully meets the practical requirements of UAV onboard deployment.By integrating multi-spectral information from the FLAME 2 dataset with the above targeted designs, this method successfully overcomes the limitations of single-modality perception, and achieves a superior balance between detection robustness and real-time performance.

Future work will focus on two directions: first, further exploring model pruning and quantization techniques, such as INT8 quantization and knowledge distillation, to adapt to micro-UAV platforms with lower computing power; second, combining UAV flight attitude data (GPS, IMU) and temporal information to study temporal modeling methods for dynamic prediction of fire spread direction.

5 Conclusion

This paper proposes Fire-YOLO26, a dual-stream forest fire detection network that achieves all-weather, end-to-end fire detection by deeply fusing RGB and thermal infrared features. The main innovations include the Thermal-guided Feature Alignment Module (T-FAM) and the Dual-modal Small Target Aware Labeling strategy (Dual-STAL). Experimental results on the FLAME 2 dataset demonstrate that Fire-YOLO26 achieves 94.6% mAP@0.5, significantly outperforming single-modality methods, while maintaining real-time inference capabilities on edge devices. This method provides an effective technical solution for early forest fire detection and has important practical application value.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

SZ: Validation, Methodology, Data curation, Conceptualization, Software, Writing – original draft. ZC: Supervision, Conceptualization, Funding acquisition, Writing – review and editing, Project administration. JX: Project administration, Writing – review and editing, Conceptualization. ZT: Data curation, Methodology, Writing – original draft, Formal Analysis, Visualization, Software. HL: Validation, Data curation, Writing – review and editing, Investigation. YF: Project administration, Supervision, Writing – review and editing. YG: Conceptualization, Supervision, Writing – review and editing, Resources. JL: Writing – review and editing, Visualization, Data curation, Investigation.

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Conflict of interest

Authors SZ, ZT, HL, YF, YG, and JL were employed by Sichuan Anxin Kechuang Technology Co., Ltd.

Authors ZC, JX and YG were employed by Sichuan Province State- owned Assets Investment Management Co., Ltd.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. During the preparation of this manuscript, the authors used Gemini to translate the initial draft and improve language readability. After using this tool, the authors reviewed and edited the content as needed and assume full responsibility for the academic accuracy and integrity of the final publication.

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References

Akhloufi, M., Couturier, A., and Castro, N. (2021). Unmanned aerial vehicles for wildland fires: sensing, perception, cooperation and assistance. *Drones* 5, 15. doi:10.3390/drones5010015

Akyon, F., Altinuc, S., and Temizel, A. (2022). "Slicing aided hyper inference and fine-tuning for small object detection," in *Proceedings of the IEEE international conference on image processing*, 966–970. doi:10.1109/ICIP46576.2022.9897990

Barmpoutis, P., Papaioannou, P., Dimitropoulos, K., and Grammalidis, N. (2020). A review on early forest fire detection systems using optical remote sensing. *Sensors* 20, 6442. doi:10.3390/s20226442

Bochkovskiy, A., Wang, C., and Liao, H. (2020). YOLOv4: optimal speed and accuracy of object detection. Available online at: https://arxiv.org/abs/2004.10934 (Accessed December 15, 2025).

Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., and Zagoruyko, S. (2020). "End-to-end object detection with transformers," in *Proceedings of the European conference on computer vision* (Cham, Switzerland: Springer), 213–229. doi:10.1007/978-3-030-58452-8_13

Chakrabarty, S. (2026). YOLO26: an analysis of NMS-Free end to end framework for real-time object detection. Available online at: https://arxiv.org/abs/2601.12882 (Accessed January 20, 2026).

Das, K., Poovvancheri, J., Flesca, S., Roberta Calidonna, C., and Chen, D. (2026). Emerging trends in wildfire detection through the lens of computer vision and wildfire emission quantification: a comprehensive survey. *IEEE Access Practical Innovations, Open Solutions* 14, 20201–20228. doi:10.1109/ACCESS.2026.3660843

de Venâncio, P., Lisboa, A., and Barbosa, A. (2022). An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices. *Neural Comput. Appl.* 34, 15349–15368. doi:10.1007/s00521-022-07467-z

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., et al. (2021). "An image is worth 16x16 words: transformers for image recognition at scale," in *International conference on learning representations*.

Frizzi, S., Kaabi, R., Bouchouicha, M., Ginoux, J.-M., Moreau, E., and Fnaiech, F. (2016). "Convolutional neural network for video fire and smoke detection," in *Iecon 2016 - 42nd annual conference of the IEEE industrial electronics society*, 877–882. doi:10.1109/IECON.2016.7793196

Hopkins, B., O'Neill, L., Afghah, F., Razi, A., Rowell, E., Watts, A., et al. (2022). FLAME 2: fire detection and modeLing: aerial multi-spectral imagE dataset. doi:10.21227/swyw-6j78

Hu, M., Ren, Y., and Chai, H. (2022). "Forest fire detection based on improved YOLOv5," in *Proceedings of the 2021 4th international conference on artificial intelligence and pattern recognition (AIPR '21)* (New York, NY, USA: Association for Computing Machinery), 172–178. doi:10.1145/3488933.3488985

Hu, J., Lin, D., and Liu, Q. (2025). A systematic review of UAV structure and monitoring models for forest fire detection. *Highlights Sci. Eng. Technol.* 134, 81–87. doi:10.54097/tdbyh330

Jocher, G., and Qiu, J. (2024). *Ultralytics YOLO11*.

Jocher, G., Chaurasia, A., and Qiu, J. (2023). *Ultralytics yolov8*.

Jordan, K., Jin, Y., Boza, V., Jiacheng, Y., Cesista, F., Newhouse, L., et al. (2024). Muon: an optimizer for hidden layers in neural networks

Lin, T., Goyal, P., Girshick, R., He, K., and Doll'ar, P. (2017). "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2980–2988. doi:10.1109/ICCV.2017.324

Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C., et al. (2016). "SSD: single shot MultiBox detector," in *European conference on computer vision*, 21–37. doi:10.1007/978-3-319-46448-0_2

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., et al. (2021). "Swin transformer: hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 10012–10022. doi:10.1109/ICCV48922.2021.00986

Liu, K., Li, T., and Peng, D. (2025a). Aerial image object detection based on RGB-infrared multi-branch progressive fusion. *IEEE Trans. Geoscience Remote Sens.* 63, 14–54. doi:10.1109/TGRS.2025.5402714

Liu, Z., Khajavi, S. H., and Jiang, G. (2025b). *DetectiumFire: a comprehensive multi-modal dataset bridging vision and language for fire understanding*. doi:10.48550/arXiv.2511.02495

Merino, L., Caballero, F., Mart'inez-de-Dios, J., Maza, I., and Ollero, A. (2012). An unmanned aircraft system for automatic forest fire monitoring and measurement. *J. Intelligent Robotic Syst.* 65, 533–548. doi:10.1007/s10846-011-9636-7

Mohapatra, A., and Trinh, T. (2022). Early wildfire detection technologies in practice—A review. *Sustainability* 14, 12270. doi:10.3390/su141912270

Muhammad, K., Ahmad, J., Lv, Z., Bellavista, P., Yang, P., and Baik, S. W. (2019). Efficient deep CNN-Based fire detection and localization in video surveillance applications. *IEEE Trans. Syst. Man, Cybern. Syst.* 49, 1419–1434. doi:10.1109/TSMC.2018.2830099

Phillips, I. I. I. W., Shah, M., and da Vitoria Lobo, N. (2002). Flame recognition in video. *Pattern Recognit. Lett.* 23, 319–327. doi:10.1016/S0167-8655(01)00132-5

Qi, H., Qin, H., Xiang, X., Yang, C., and Tan, Y. (2026). LF-DETR: a laplacian frequency enhanced DETR for aerial RGB-infrared pedestrian detection. *Remote Sens.* 18, 531. doi:10.3390/rs18030531

Redmon, J., and Farhadi, A. (2017). "YOLO9000: better, faster, stronger," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 7263–7271. doi:10.1109/CVPR.2017.690

Redmon, J., and Farhadi, A. (2018). YOLOv3: an incremental improvement. Available online at: https://arxiv.org/abs/1804.02767 (Accessed December 18, 2025).

Redmon, J., Divvala, S., Girshick, R., and Farhadi, A. (2016). "You only look once: unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 779–788. doi:10.1109/CVPR.2016.91

Ren, S., He, K., Girshick, R., and Sun, J. (2015). "Faster R-CNN: towards real-time object detection with region proposal networks," in *Advances in neural information processing systems*, 91–99.

Sapkota, R., Cheppally, R., Sharda, A., and Karkee, M. (2025). *YOLO26: key architectural enhancements and performance benchmarking for real-time object detection*. Available online at: https://arxiv.org/abs/2509.25164 (Accessed January 10, 2026).

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). "Grad-CAM: visual explanations from deep networks *via* gradient-based localization," in *2017 IEEE international conference on computer vision (ICCV)*, 618–626. doi:10.1109/ICCV.2017.74

Shamsoshoara, A., Afghah, F., Razi, A., Zheng, L., Fulé, P. Z., and Blasch, E. (2021). Aerial imagery pile burn detection using deep learning: the FLAME dataset. *Comput. Netw.* 193, 108001. doi:10.1016/j.comnet.2021.108001

Sun, Y., Jiang, L., Pan, J., Sheng, S., and Hao, L. (2023). A satellite imagery smoke detection framework based on the mahalanobis distance for early fire identification and positioning. *Int. J. Appl. Earth Observation Geoinformation* 118, 103257. doi:10.1016/j.jag.2023.103257

Tan, M., Pang, R., and Le, Q. (2020). "EfficientDet: scalable and efficient object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 10781–10790. doi:10.1109/CVPR42600.2020.01028

Tian, Y., Ye, Q., and Doermann, D. (2025). *YOLOv12: attention-Centric real-time object detectors*. Available online at: https://arxiv.org/abs/2502.12524 (Accessed February 5, 2026).

Tong, K., Wu, Y., and Zhou, F. (2020). Recent advances in small object detection based on deep learning: a review. *Image Vis. Comput.* 97, 103910. doi:10.1016/j.imavis.2020.103910

Toreyin, B., Dedeoglu, Y., and Cetin, A. (2005). "Flame detection in video using hidden markov models," in *Proceedings of the IEEE international conference on image processing*, 1230–1233. doi:10.1109/ICIP.2005.1530570

Vasconcelos, R. N., Franca Rocha, W. J. S., Costa, D. P., Duverger, S. G., de Santana, M. M. M., Cambui, E. C. B., et al. (2024). Fire detection with deep learning: a comprehensive review. *Land* 13, 1696. doi:10.3390/land13101696

Wang, C., Liao, H., Wu, Y., Chen, P., Hsieh, J., and Yeh, I. (2020). "CSPNet: a new backbone that can enhance learning capability of CNN," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, 1571–1580. doi:10.1109/CVPRW50498.2020.00203

Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., et al. (2024). YOLOv10: real-Time end-to-end object detection. Available online at: https://arxiv.org/abs/2405.14458 (Accessed January 25, 2026).

Wang, H., Zhang, Y., and Zhu, C. (2025). YOLO-LFD: a lightweight and fast model for forest fire detection. *Comput. Mater. and Continua* 82, 3399–3417. doi:10.32604/cmc.2024.058932

Woo, S., Park, J., Lee, J., and Kweon, I. (2018). "CBAM: convolutional block attention module," in *Proceedings of the European conference on computer vision* (Cham, Switzerland: Springer), 3–19. doi:10.1007/978-3-030-01234-2_1

World Resources Institute (2024). Global forest watch: fire monitoring. Available online at: https://www.globalforestwatch.org/ (Accessed February 12, 2026).

Xu, Y., Wang, H., Bi, Y., Nie, G., and Wang, X. (2025). MCDet: target-Aware fusion for RGB-T fire detection. *Forests* 16, 1088. doi:10.3390/f16071088

Yun, S., Han, D., Oh, S., Chun, S., Choe, J., and Yoo, Y. (2019). "CutMix: regularization strategy to train strong classifiers with localizable features," in *Proceedings of the IEEE/CVF international conference on computer vision*, 6023–6032. doi:10.1109/ICCV.2019.00612

Zhang, H., Cisse, M., Dauphin, Y., and Lopez-Paz, D. (2018). "Mixup: beyond empirical risk minimization," in *Proceedings of the international conference on learning representations*.

Zhang, Y., Xu, C., Yang, W., He, G., Yu, H., Yu, L., et al. (2023). Drone-based RGBT tiny person detection. *ISPRS J. Photogrammetry Remote Sens.* 204, 61–76. doi:10.1016/j.isprsjprs.2023.08.016

Zhu, X., Su, W., Lu, L., Li, B., Wang, X., and Dai, J. (2021). *Deformable DETR: deformable transformers for end-to-end object detection*. doi:10.48550/arXiv.2010.04159